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Information Systems in GIS

SUBJECT: GEOINFORMATICS-II

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Information

- Every aspect of management in the modern age relies heavily on information to thrive.
- Nothing moves without information and it is generally believed that information is power and that he who has it has power.
- It is an important resource needed to develop other resources.
- Changing circumstances and environments have necessitated the need for the proper dissemination of information at various levels of management.

Information

- The concept of information in an organizational sense is more complex and difficult than the frequent use of this common word would suggest.
- Every society, no doubt, is an information society and every organization is an information organization.
- Therefore, information is a basic resource like materials, money and personnel.
- Information can be considered either as an abstract concept(ideas) or as a commodity, usually in the form of letters and reports.

Information System

- **Information system**, an integrated set of components for collecting, storing, and processing data and for providing information, knowledge, and digital products. Business firms and other organizations rely on information systems to carry out and manage their operations, interact with their customers and suppliers, and compete in the marketplace.
- Information systems are used to run inter organizational supply chains and electronic markets. For instance, corporations use information systems to process financial accounts, to manage their human resources, and to reach their potential customers with online promotions.

Information System

- Many major companies are built entirely around information systems. These include eBay, a largely auction marketplace; Amazon, an expanding electronic mall and provider of cloud computing services; Alibaba, a business-to-business e-marketplace; and Google, a search engine company that derives most of its revenue from keyword advertising on Internet searches.
- Governments deploy information systems to provide services cost-effectively to citizens.
- Digital goods—such as electronic books, video products, and software—and online services, such as gaming and social networking, are delivered with information systems. Individuals rely on information systems, generally Internet-based, for conducting much of their personal lives: for socializing, study, shopping, banking, and entertainment.

Information System

- Any organized combination of
 - People
 - Hardware
 - Communication networks
 - Software
 - Data resources
 - Policies and procedures that stores retrieves, transforms, and disseminates information in an organization.

Information System



Components of Information System

1. **Computer hardware.** This is the physical technology that works with information. ...
2. **Computer software.** The hardware needs to know what to do, and that is the role of software. ...
3. **Telecommunications.** ...
4. **Databases and data warehouses.** ...
5. **Human resources and procedures.**

THE COMPONENTS OF INFORMATION SYSTEMS

SYSTEM CONCEPTS: A FOUNDATION

Technology

Applications

Development

Management

WHAT IS A SYSTEM?

Input

Processing

Output

FEEDBACK AND CONTROL

The system concept becomes even more useful by including two additional elements:

Feedback

Control

Feedback is data about the performance of a system.

For example:

data about sales performance is feedback to a sales manager.

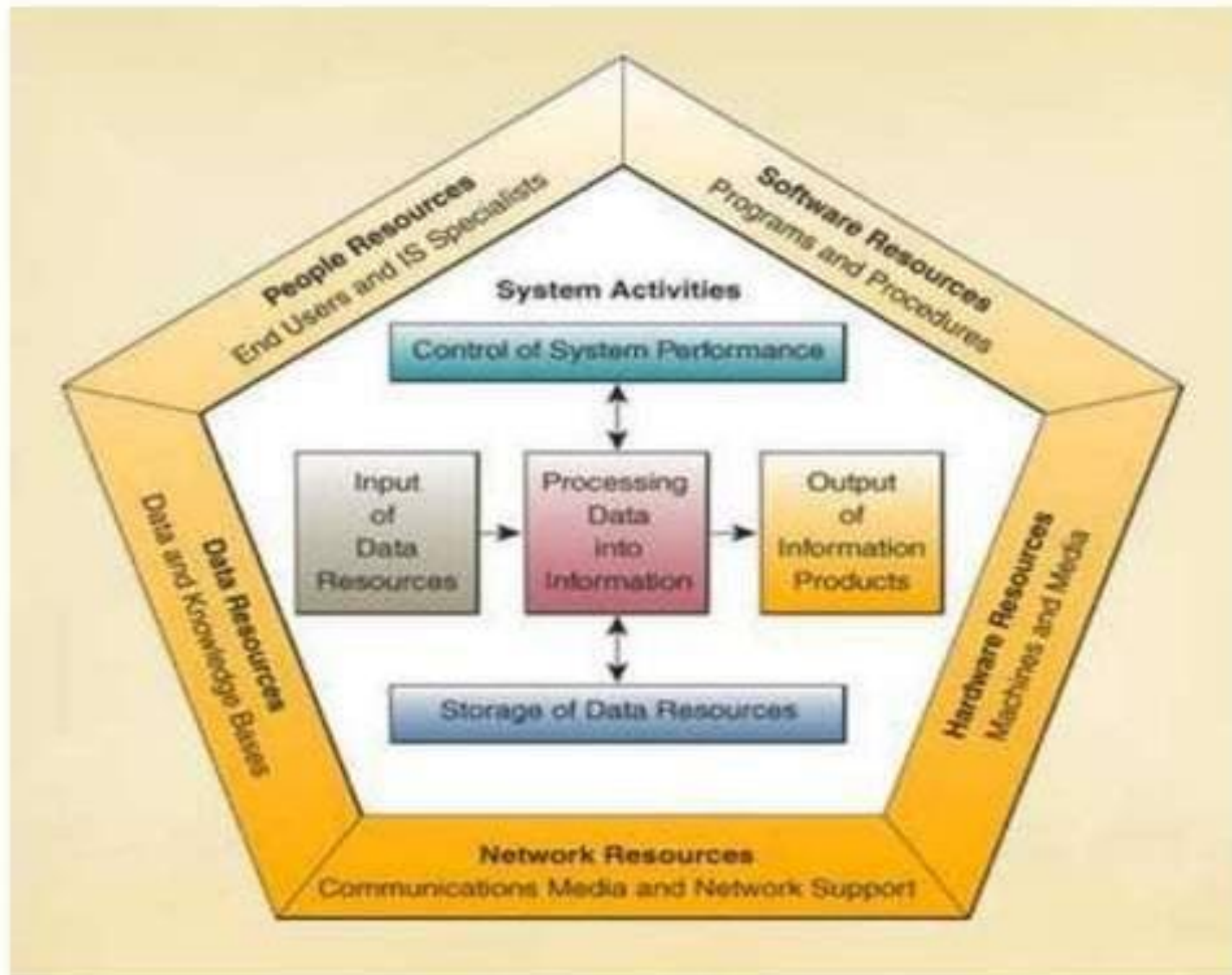
Data about the speed, attitude, and direction of an aircraft is feedback to the aircraft's pilot or autopilot.

Control involves monitoring and evaluating feedback to determine whether a system is moving toward the achievement of its goal.

The control function then makes the necessary adjustments to a system's input and processing components to ensure that it produces proper output.

For example: An airline pilot, makes minute adjustments after evaluating the feedback from the instruments to ensure the plane is exactly where the pilot wants it to be.

Component of IS



➤ INFORMATION SYSTEM RESOURCES

PEOPLE RESOURCES

- SPECIALISTS
- END USERS

HARDWARE RESOURCES

- MACHINES
- MEDIA

SOFTWARE RESOURCES

- PROGRAMS
- PROCEDURES

DATA RESOURCES

- PRODUCT DESCRIPTIONS, EMPLOYEE FILES

NETWORK RESOURCES

- COMMUNICATIONS MEDIA, CONTROL SOFTWARE

INFORMATION PRODUCTS

- MANAGEMENT REPORTS AND BUSINESS DOCUMENTS

Functional elements of IS

- It comprises the following functional elements which relate to the organization and its environments:
 1. **Perception**– initial entry of data whether captured or generated, into the organization;
 2. **Recording**– physical capture of data;
 3. **Processing**– transformation according to the “specific” needs of the organization;
 4. **Transmission**– the flows which occur in an information system;
 5. **Storage**– presupposes some expected future use;
 6. **Retrieval**– search for recorded data;
 7. **Presentation**– reporting, communication; and
 8. **Decision making**– a controversial inclusion, except to the extent that the information system engages in decision making that concerns itself.

Types of IS

The most important computer-based information systems as follows:

1. Information retrieval system (IR);
2. Question-answering system;
3. Database system (DBS);
4. Management information system (MIS);
5. Decision support system (DSS).

Application of IS

Applications of Information Systems



Recognizing IS

As a business professional, we should be able to recognize the fundamental components of IS we encounter in the real world. This means that you should be able to identify:

The people, hardware, software, data, and network resources they use.

The types of information products they produce.

The way they perform input, processing, output, storage, and control activities.

How a Person receive information

1. Visually → through sight.

Ex: receive information by seeing the print in book.

2. Aurally → through hearing.

Ex: hear the whistle of a train.

3. Tactilely → through touch

Ex: feeling the humidity in the air.

4. Olfactory → through smell

Ex: smelling the coffee in the morning.

5. Gustatorily → through taste

Ex: tasting to see whether the milk is fresh or not .

Some of the information activities we receive involve transforming it from one format to another.

Eg: Studying , building a house.

How a Person Use information

- Information devices and services provide to the individual information that is used for a variety of purpose in the conduct of daily life.!

1. Make decisions!

Information about course in university webpage (enroll or no) !

2. Resolve uncertainty !

Ex: looking at the clock while studying for exam, you are trying to resolve uncertainty of wither to stop reading and go to class or not.!

3. Solve problems

Various Sphere of Information in Society

1. **Economic sphere:** Information is used daily to influence behavior,
ex : advertising , oil. !
2. **Political sphere:** to influence behavior of citizens (voting, government response)!
3. **In education sector:** teaching, learning, and research !
4. **In health sector:** for patient care and drug prescription !
5. **In military and security:** for war, fighting and protection from crime

Information Society

- Webster (2002) who proposes examining a society on the basis of economic, occupational, spatial, technology and cultural criteria to determine the extent to which an information society exists. (Lester & Koehler, 2007)

Types of IS

- **The oral society**
 - •Where most people reality and life was local
Knowing a very little about other places, people
and societies
 - Society and societal institutional were oral
Information was stored in human memory.
 - Human memory was the major way of
transporting information across time and space
- Sources of information and authority were old
people those who had more information stored
in their memory (Lester & Koehler, 2007)

- **Print Society**

It provided an accessible way to store information, Information container, can gave access to information to much larger body of people

That led to change the nature of the information transmitted which helps in increasing in mutual access to information, the sharing of information and ideas, changed the reality of society (Lester & Koehler, 2007)

Type of IS

- **Beyond print to electronic society**

Telegraph and telephone

Telegraph made the transmission of messages possible across space and time from one individual to another. Telephone where individuals can share information through individual interactive oral communication across space and time

Radio

Which support sharing of information in large scale (Lester & Koehler, 2007)

Television

It changes the sense of community through provision of information, extending the community of the individual beyond the physical location

Computer and the Internet

Which support recording, storing and distribution of information in digital format by using combination of networked digital infrastructure (Lester & Koehler, 2007)

Information System and Organization

- The rapid evolution of computer technology is expanding .
- man's desire to obtain computer assistance in solving more and more complex problems: problems which were considered solely in the domain of man's intuitive and judgmental processes, particularly in organizations, a few years ago.
- Information systems are becoming of ever greater interest in progressive and dynamic organizations.
- The need to obtain access conveniently, quickly and economically makes it imperative to devise procedures for the creation, management and utilization of databases in organizations.
- Management information and information systems, in particular those related to effective decision-making processes in a organization.

IS and Organization

- Information systems can give you a competitive advantage and provide the data you need to make faster, smarter business decisions. Depending on your needs, you can opt for transaction processing systems, knowledge management systems, decision support systems and more.
- When choosing one, consider your budget, industry and business size. Look for an information system that aligns with your goals and can streamline your day-to-day operations.
- Business intelligence (BI) systems, for instance, can turn data into valuable insights.
- IS allows for faster, more accurate reporting, better business decisions and more efficient resource allocation.

ERP IS in Organization

- Another major benefit is data visualization, which enables analysts to interpret large amounts of information, predict future events and find patterns in historical data.
- Organizations can also use enterprise resource planning (ERP) software to collect, manage and analyze data across different areas, from manufacturing to finance and accounting.
- This type of information system consists of multiple applications that provide a 360-degree view of business operations. Net Suite ERP, PeopleSoft.
- ERP provides actionable insights and helps you decide on the next steps.
- It also makes it easier to achieve regulatory compliance, increase data security and share information between departments.
- In the long run, ERP software can reduce operational costs, improve collaboration and boost your revenue.
- Nearly half of the companies that implement this system report major benefits within six months.

IS for a Business Organization

1. **Support the Business Process :** Treats inputs as a request from the **customer** and outputs as services to customer. Supports current operations and use the system to influence further way of working.
2. **Support Operation of a Business Organization:** MIS supports **operations of a** business organization by giving timely information, maintenance and enhancement which provides flexibility in the operation of organizations.
3. **To Support Decision Making:** MIS supports the **decision making by employee in** their daily operations. MIS also supports managers in decision making to meet the goals and objectives of the organization. Different mathematical models and IT tools are used for the purpose evolving strategies to meet competitive needs.
4. **Strategies for an Organization:** Today each business is running in a **competitive** market. MIS supports the organization to evolve appropriate strategies for the business to assented in a competitive environment.

Constraint of IS

- Belief that a computerized MIS can solve all problems.
- Developed without streamlining the TPS.
- It is conceived as data processing and not as information system.
- MIS does not give perfect information to all users.
- Lack of administrative discipline in following standardized systems and procedures, wrong coding can result in incomplete and incorrect information.
- MIS does not provide information needed by managers but that what the function calls for.
- MIS does not meet critical and key factors such as response to query on database etc.

Types of GIS Information Systems

1.Desktop GIS:Description: Software applications that run on personal computers. These are used for creating, analyzing, and visualizing spatial data.

1. **Examples:** ESRI ArcGIS Desktop, QGIS, MapInfo Professional.
2. **Use Cases:** Detailed analysis, cartographic design, and spatial modeling.

Desktop GIS (Geographic Information Systems) refers to GIS software applications that are installed and run on personal computers or workstations. These applications are used for capturing, analyzing, managing, and visualizing spatial and geographic data. Desktop GIS provides powerful tools for spatial analysis and mapping, making it a key component in many organizations' geospatial workflows.

Popular Desktop GIS Software

1. Esri ArcGIS Pro:

1. **Overview:** A powerful desktop GIS application developed by Esri, offering advanced spatial analysis, 3D visualization, and data management capabilities.
2. **Features:** Includes advanced mapping tools, data editing, and integration with ArcGIS Online and ArcGIS Enterprise.

2. QGIS (Quantum GIS):

1. **Overview:** An open-source GIS application that provides a wide range of tools for spatial analysis, data visualization, and cartography.
2. **Features:** Includes a rich set of plugins, support for multiple data formats, and extensive community support.

3. MapInfo Professional:

1. **Overview:** A desktop GIS application developed by Pitney Bowes Software, known for its mapping and spatial analysis capabilities.
2. **Features:** Offers tools for data visualization, spatial analysis, and geographic data management.

Web GIS:

GIS applications that are accessed via a web browser. They enable users to interact with spatial data over the internet.

Examples: ESRI ArcGIS Online, Google Maps, OpenStreetMap.

Use Cases: Public mapping services, data sharing, and collaborative analysis.

Web GIS, or Web Geographic Information Systems, refers to the integration and use of geographic information system (GIS) technologies and data over the internet. It allows users to access, analyze, and interact with spatial data and mapping tools through web browsers or mobile apps. Here's a breakdown of the concept:

1. Geographic Information Systems (GIS): GIS is a framework for gathering, managing, and analyzing spatial and geographic data. It involves hardware, software, data, and people working together to understand and interpret location-based information.

2. Web-Based Access: Web GIS extends the functionality of traditional GIS by making geographic information and tools accessible online. This means users don't need specialized software installed on their devices; they can interact with maps and spatial data directly through a web browser.

Key Components of Web GIS

- 1. Web Servers:** Host GIS applications and spatial data. They manage user requests and deliver map tiles or data.
- 2. Web Mapping Applications:** Provide an interface for users to interact with maps. These applications can include features such as zooming, panning, querying, and overlaying different types of data.
- 3. Geospatial Data:** Includes various types of data such as satellite imagery, vector data (e.g., points, lines, and polygons), and tabular data with geographic references.
- 4. Web Services:** Allow the sharing of GIS data and functionality over the internet. Common standards include Web Map Service (WMS) and Web Feature Service (WFS), which enable the retrieval of map images and geographic features respectively.

Mobile GIS:

GIS applications designed for mobile devices, such as smartphones and tablets. They are used for field data collection and real-time data access.

- 1. Examples:** ESRI Collector for ArcGIS, Survey123, Locus Map.
- 2. Use Cases:** Field surveys, navigation, and real-time data gathering.

Mobile GIS (Geographic Information Systems) refers to the use of GIS technology on mobile devices such as smartphones and tablets to access, collect, and analyze geographic data. This concept extends the capabilities of traditional GIS to mobile platforms, providing users with the flexibility to interact with spatial data and mapping tools while on the move. Here's a detailed explanation:

Key Concepts of Mobile GIS

1.Mobile Devices: Mobile GIS leverages smartphones, tablets, and other handheld devices equipped with GPS, sensors, and internet connectivity to perform GIS functions. These devices enable users to access maps, collect field data, and perform spatial analyses directly from their location.

GIS Functionality on Mobile Platforms:

1. **Mapping and Navigation:** Users can view and interact with maps, get directions, and navigate through geographic locations.
2. **Data Collection:** Mobile GIS allows users to capture spatial data in the field using GPS coordinates, camera images, and forms. This includes collecting information on points of interest, environmental conditions, or infrastructure.
3. **Real-Time Updates:** Data collected through mobile GIS can be synchronized with central databases in real-time or near-real-time, ensuring that the information is current and accurate.

Key Features:

- **Location-Based Services:** Utilize GPS and other location technologies to provide location-specific data and services, such as finding nearby points of interest or tracking movement.
- **Offline Capabilities:** Many mobile GIS applications support offline functionality, allowing users to access maps and collect data without a continuous internet connection. Data can be synchronized later when connectivity is available.
- **User Interfaces:** Mobile GIS applications are designed with user-friendly interfaces optimized for touchscreens, making it easy for users to interact with maps, input data, and perform tasks.

Enterprise GIS:

- 1. Description:** Comprehensive systems that integrate GIS capabilities across an organization. They often involve a centralized database and support multiple users with varying roles.
- 2. Examples:** ESRI ArcGIS Enterprise, Bentley Systems.
- 3. Use Cases:** Large-scale data management, enterprise-wide analysis, and decision support.

ArcGIS Enterprise is a comprehensive and scalable GIS platform developed by Esri, designed to manage, analyze, and share geographic information across an organization. It provides the infrastructure needed for deploying a full suite of GIS capabilities within an organization's IT environment, whether on-premises, in the cloud, or in a hybrid setup. Here's an overview of its key components and features:

Key Components of ArcGIS Enterprise

1. Portal for ArcGIS:

- 1. Function:** Acts as the central hub for accessing and sharing GIS content and applications within the organization. It provides a web-based interface where users can view, create, and manage maps, apps, and other spatial data.
- 2. Features:** Includes customizable web maps, dashboards, and data visualization tools. It also supports user roles and permissions to control access to GIS resources.

2. ArcGIS Server:

- 1. Function:** Hosts and serves GIS services, such as map services, feature services, and geocoding services. It enables users to publish and access spatial data and analytical tools through web applications and other client applications.
- 2. Features:** Supports high-performance map rendering, spatial analysis, and data management. It can scale to handle large datasets and multiple concurrent users.

3. ArcGIS Data Store:

1. **Function:** Provides a centralized data repository for storing and managing spatial data used by ArcGIS Server and Portal for ArcGIS. It includes multiple types of data stores, such as relational, tile cache, and spatiotemporal.
2. **Features:** Ensures efficient data management and retrieval, supporting various data types and storage needs.

4. ArcGIS Enterprise Sites:

1. **Function:** Allows organizations to create and manage customized websites and web applications tailored to specific needs or audiences.
2. **Features:** Includes tools for building web maps, applications, and other GIS resources that can be accessed through a branded and user-friendly interface.

5. ArcGIS Pro:

1. **Function:** Although not part of the core ArcGIS Enterprise components, ArcGIS Pro is often used in conjunction with it. It is a desktop GIS application for advanced spatial analysis, mapping, and 3D visualization.
2. **Features:** Provides a comprehensive suite of tools for data editing, analysis, and visualization, with integration capabilities to ArcGIS Enterprise for sharing and collaboration.

CLOUD GIS:

GIS systems that leverage cloud computing infrastructure to store, process, and analyze spatial data.

1. **Examples:** Amazon Web Services (AWS) with GIS applications, ESRI ArcGIS Online.
2. **Use Cases:** Scalability for large datasets, collaborative projects, and distributed data management.

Cloud GIS refers to the use of Geographic Information Systems (GIS) technologies and spatial data in a cloud computing environment. This approach leverages cloud infrastructure to provide scalable, flexible, and accessible GIS capabilities over the internet.

Key Concepts of Cloud GIS

1. Cloud Computing:

1. **Definition:** Cloud computing delivers computing services—including servers, storage, databases, networking, software, and more—over the internet (the cloud). This eliminates the need for local hardware and software installations.
2. **Benefits:** Offers on-demand resource provisioning, scalability, pay-as-you-go pricing, and remote access.

2. Cloud GIS:

1. **Definition:** Cloud GIS utilizes cloud computing to deliver GIS applications and services. It enables users to store, analyze, and share geographic data using cloud-based platforms rather than traditional on-premises systems.
2. **Components:** Includes cloud-based GIS applications, data storage, and processing services.

KEY COMPONENTS OF CLOUD GIS

1. Cloud-Based GIS Platforms:

1. **Examples:** ArcGIS Online (by Esri), Google Earth Engine, Microsoft Azure Maps, and AWS GIS Services. These platforms offer web-based access to mapping, spatial analysis, and data management tools.

2. Web Map Services (WMS) and Web Feature Services (WFS):

1. **Definition:** These are standard protocols for serving geospatial data over the internet. WMS provides map images, while WFS provides raw geographic features.
2. **Usage:** These services enable the integration of external spatial data into cloud GIS applications.

3. APIs and SDKs:

1. **Definition:** Application Programming Interfaces (APIs) and Software Development Kits (SDKs) allow developers to create custom GIS applications and integrate cloud GIS capabilities into existing systems.
2. **Examples:** Esri's ArcGIS API for JavaScript, Google Maps API.

4. Data Integration and Interoperability:

1. Cloud GIS platforms often support integration with various data sources, including databases, real-time data feeds, and other cloud services, enhancing data interoperability and usability.

Spatial Data Infrastructure (SDI):

Frameworks that support the collection, sharing, and use of spatial data across different organizations and sectors.

- 1. Examples:** INSPIRE (Infrastructure for Spatial Information in Europe), National Spatial Data Infrastructure (NSDI) in various countries.
- 2. Use Cases:** Government data sharing, cross-agency coordination, and international spatial data collaboration.

Spatial Data Infrastructure (SDI) refers to a framework that facilitates the sharing, management, and utilization of spatial data across different organizations and users. It provides the technical, organizational, and policy framework needed to enable effective and efficient access to and use of geographic and spatial information. SDI integrates various components to ensure that spatial data is interoperable, accessible, and usable across different systems and applications.

Key Components of Spatial Data Infrastructure

1.Data:

- 1. Definition:** Geographic and spatial data, including maps, satellite imagery, geographic coordinates, and related metadata.
- 2. Types:** Can include vector data (e.g., points, lines, polygons), raster data (e.g., aerial imagery, digital elevation models), and tabular data with geographic references.

2. Standards:

1. **Definition:** Guidelines and specifications that ensure data compatibility, consistency, and interoperability.
2. **Examples:** Standards like the Open Geospatial Consortium (OGC) standards (e.g., Web Map Service (WMS), Web Feature Service (WFS)), and metadata standards like ISO 19115.

3. Technology:

1. **Definition:** Tools and systems used to manage, process, and disseminate spatial data.
2. **Examples:** Geographic Information Systems (GIS), databases, web services, and spatial data servers.

4. Policies and Governance:

1. **Definition:** Rules and procedures that govern the use, sharing, and management of spatial data.
2. **Examples:** Data sharing agreements, privacy policies, data stewardship guidelines, and compliance with legal and regulatory requirements.

5. Human Resources:

1. **Definition:** The people involved in developing, managing, and utilizing spatial data.
2. **Examples:** Data producers, GIS specialists, policy makers, and end-users.

6. Infrastructure:

1. **Definition:** The underlying hardware and network systems that support spatial data management and access.
2. **Examples:** Servers, cloud services, and network infrastructure.

THANK YOU